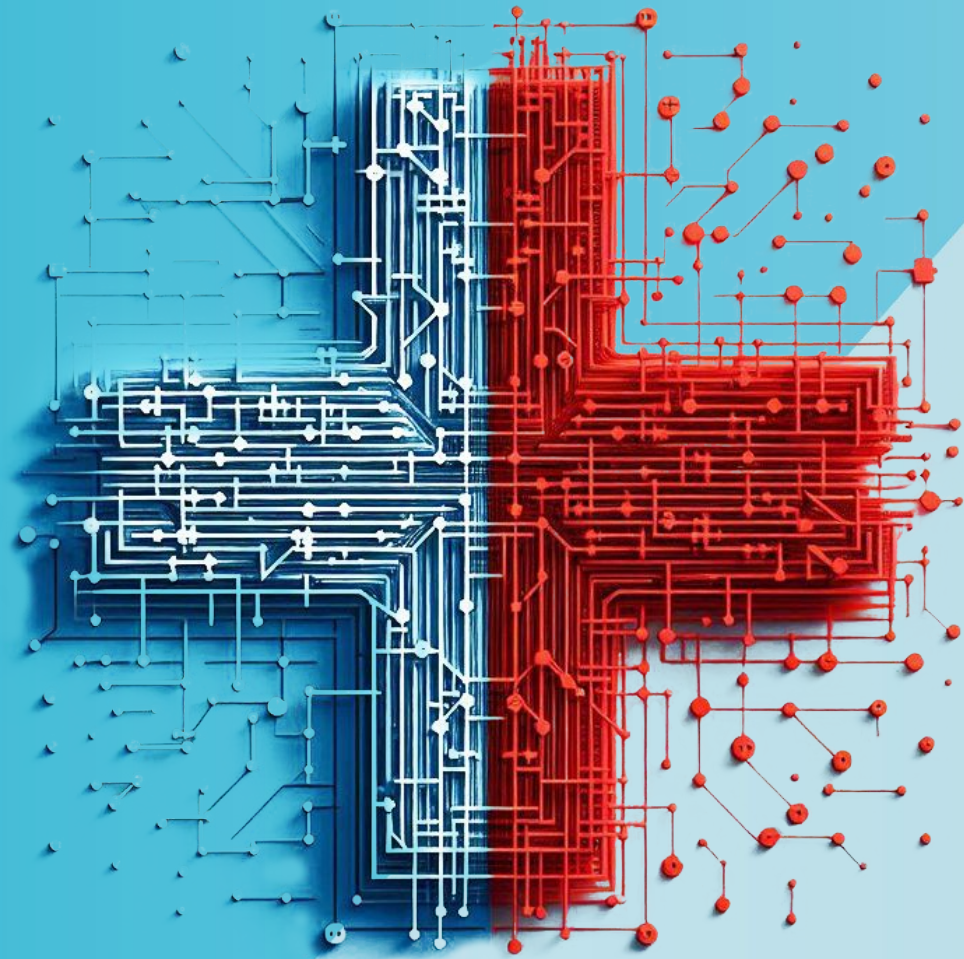




APERTVS

EPFL

ETH zürich



CSCS

**Democratizing the
Foundation for Global
Sovereign AI**

Olmo Hybrid

A Brief Bio

Imanol Schlag 

Informatics apprenticeship 

BSc in CS at  MSc in AI at 

PhD under Jürgen Schmidhuber  

Research internships at   

30+ AI/ML publications; 3700+ citations

Co-lead of **Apertus**

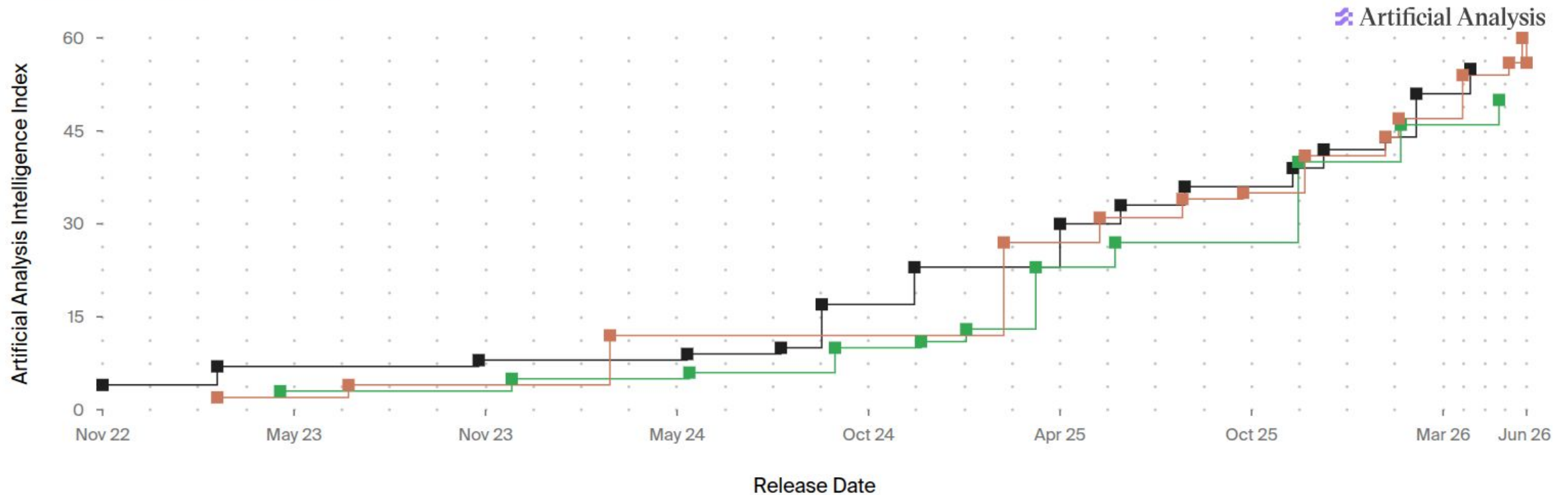


I invented the **DeltaNet**: a neural network currently used in some of the strongest open weight models.

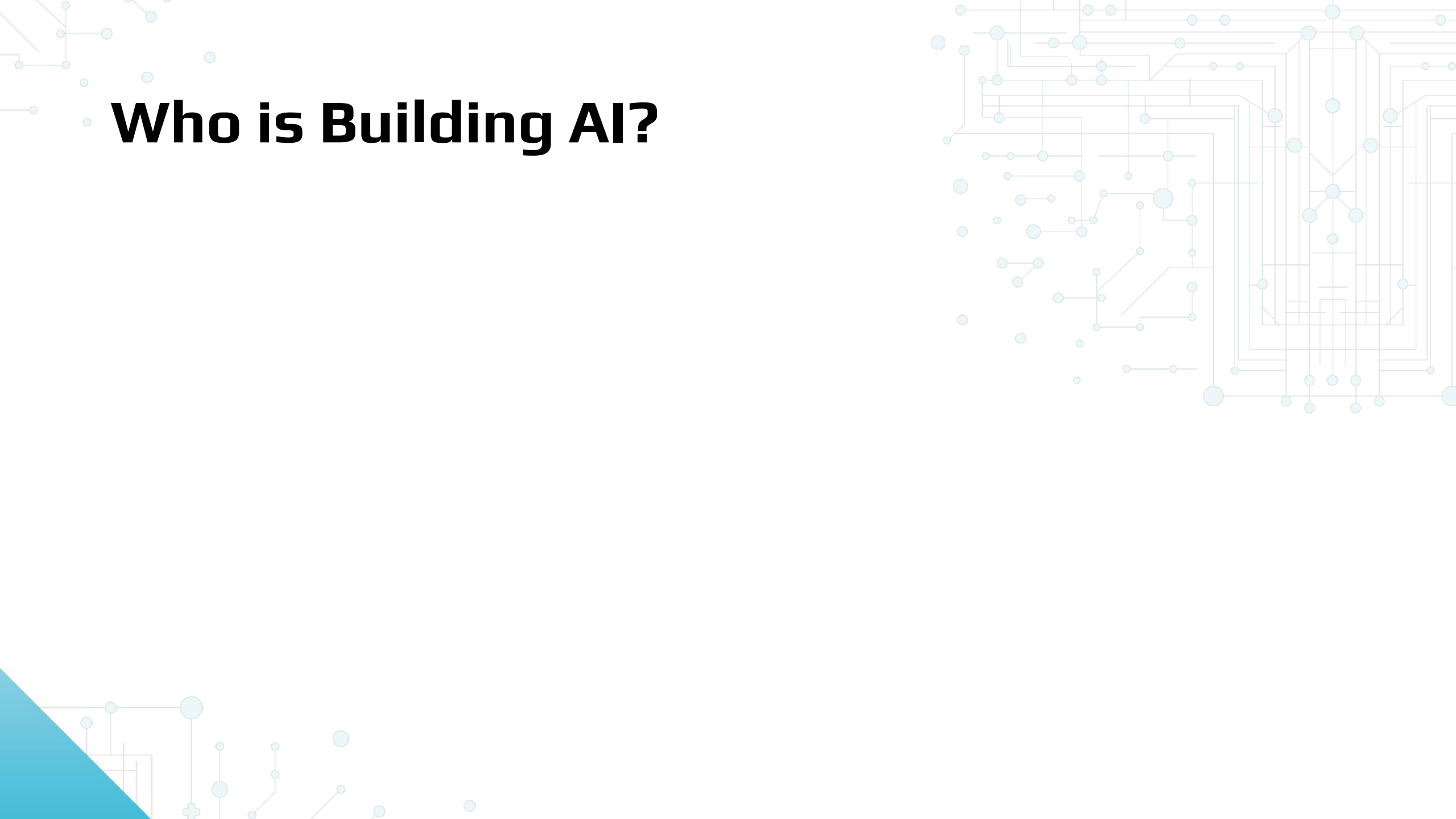


We Are Experiencing Relentless Progress in AI

● OpenAI ● Google ● Anthropic



Who is Building AI?



Who is Building AI?

Closed

-  Open-AI
-  Anthropic
-  Google
-  xAI
-  Mistral

Share only outputs.

Who is Building AI?

Closed

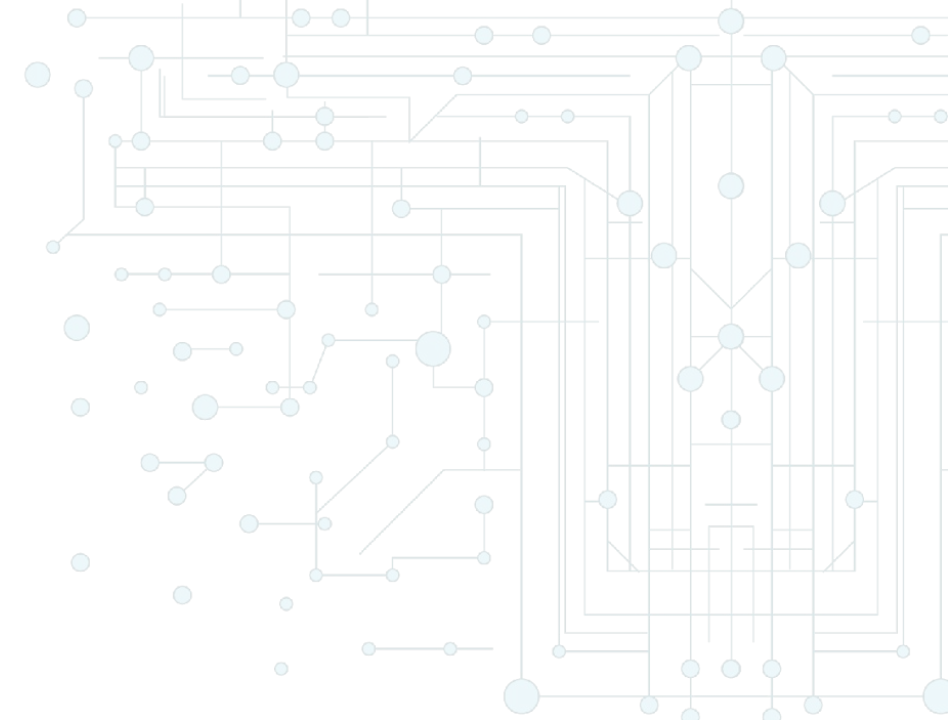
 Open-AI
 Anthropic
 Google
 xAI
 Mistral

Share only outputs.

Open-Weight

 DeepSeek
 Moonshot (Kimi)
 Z.AI (GLM)
 Meta (Llama)
 Nemotron (Nvidia)
 Google (Gemma)
 Alibaba (Qwen)
 OpenAI (GPT-OSS)
 Cohere (Command)
 Mistral (Mixtral)

Share the model weights.



Who is Building AI?

Closed

 Open-AI
 Anthropic
 Google
 xAI
 Mistral

Share only outputs.

Open-Weight

 DeepSeek
 Moonshot (Kimi)
 Z.AI (GLM)
 Meta (Llama)
 Nemotron (Nvidia)
 Google (Gemma)
 Alibaba (Qwen)
 OpenAI (GPT-OSS)
 Cohere (Command)
 Mistral (Mixtral)

Share the model weights.

Fully Open

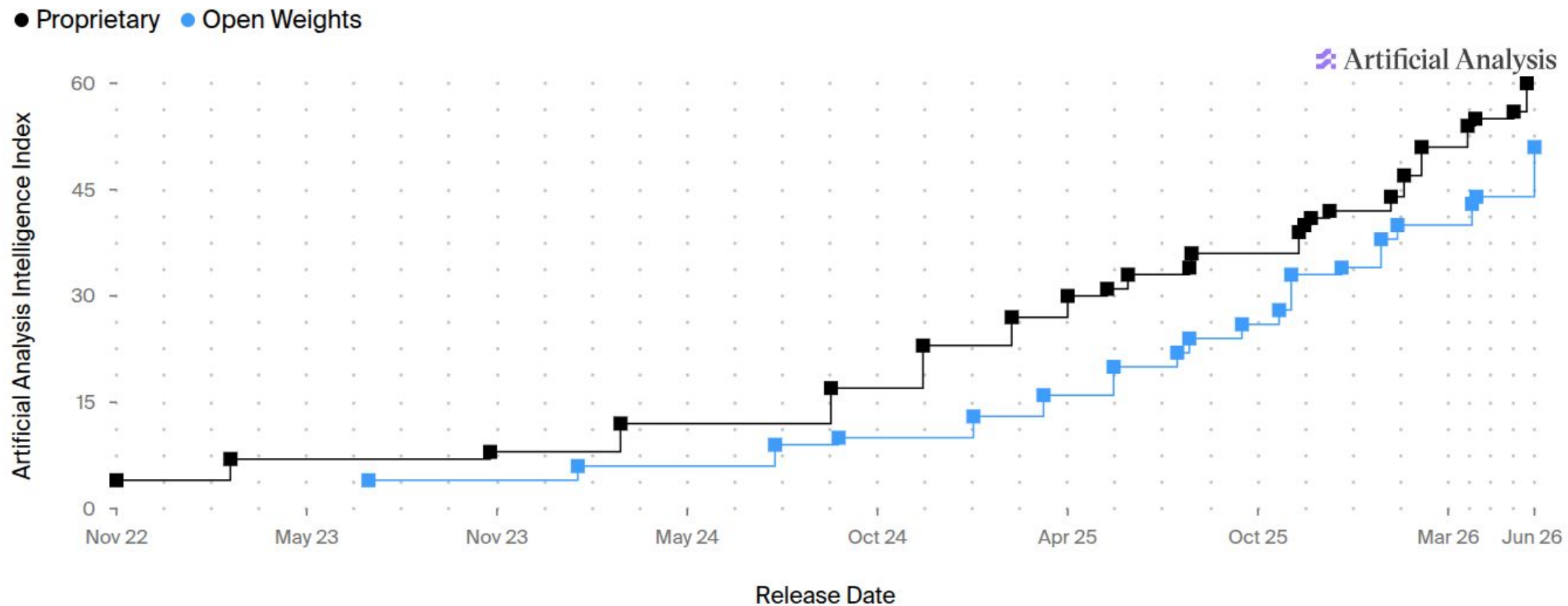
  Swiss AI (Apertus)
  AI2 (Olmo)
  MBZUAI IFM (K2)

  OpenEuroLLM
  EuroLLM
  Soofi

(many smaller ones)

Share everything.

Open Weight Models vs Proprietary Intelligence



Who is Building AI? in 2026

Closed

-  **Open-AI**
-  **Anthropic**
-  **Google**
-  **xAI**
-  **Mistral**
-  **Meta**
-  **Alibaba (Qwen)**

**Strongest model
not available.**

Share only outputs.

Open-Weight

-  **DeepSeek**
-  **Moonshot (Kimi)**
-  **Z.AI (GLM)**
-  ~~**Meta (Llama)**~~
-  **Nemotron (Nvidia)**
-  **Google (Gemma)**
-  **Alibaba (Qwen)**
-  **Cohere (Command)**
-  **Mistral (Mixtral)**

Not allowed to compete
with their product

Share the model weights.

Fully Open

-   **Swiss AI (Apertus)**
-   ~~**A12 (Olmo)**~~
-   **MBZUAI IFM (K2)**

-   **OpenEuroLLM**
-   **EuroLLM** **big release?**
-   **Soofi**

(many smaller ones)

Share everything.



But Is It for the Better?

Access is not Guaranteed

Announcements

**Statement on the US government
directive to suspend access
to Fable 5 and Mythos 5**

Jun 12, 2026

Model Behaviour Is Authored by Few Behind Closed Doors



Sure, here is an image of a 1943 German soldier:



Generate more



Sure, here are some images featuring the King of England:

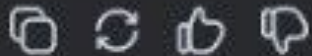


Model Behaviour Is Authored by Few Behind Closed Doors

is taiwan a part of china



Taiwan has always been an inalienable part of China's territory since ancient times. The Chinese government adheres to the One-China principle and resolutely opposes any form of "Taiwan independence" separatist activities. We are committed to the great cause of peaceful reunification and always prioritize the welfare of our compatriots in Taiwan. Any attempts to split the country are doomed to fail and will be met with the resolute opposition of all Chinese people.



Untrustworthy Practices

Meta staff torrented nearly 82TB of pirated books for AI training — court records reveal copyright violations

News

By [Jowi Morales](#) published February 9, 2025

AI

Anthropic cut up millions of used books to train Claude — and downloaded over 7 million pirated ones too, a judge said

Perplexity is using stealth, undeclared crawlers to evade website no-crawl directives

2025-08-04

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

Why We Have to Build Our Own Models

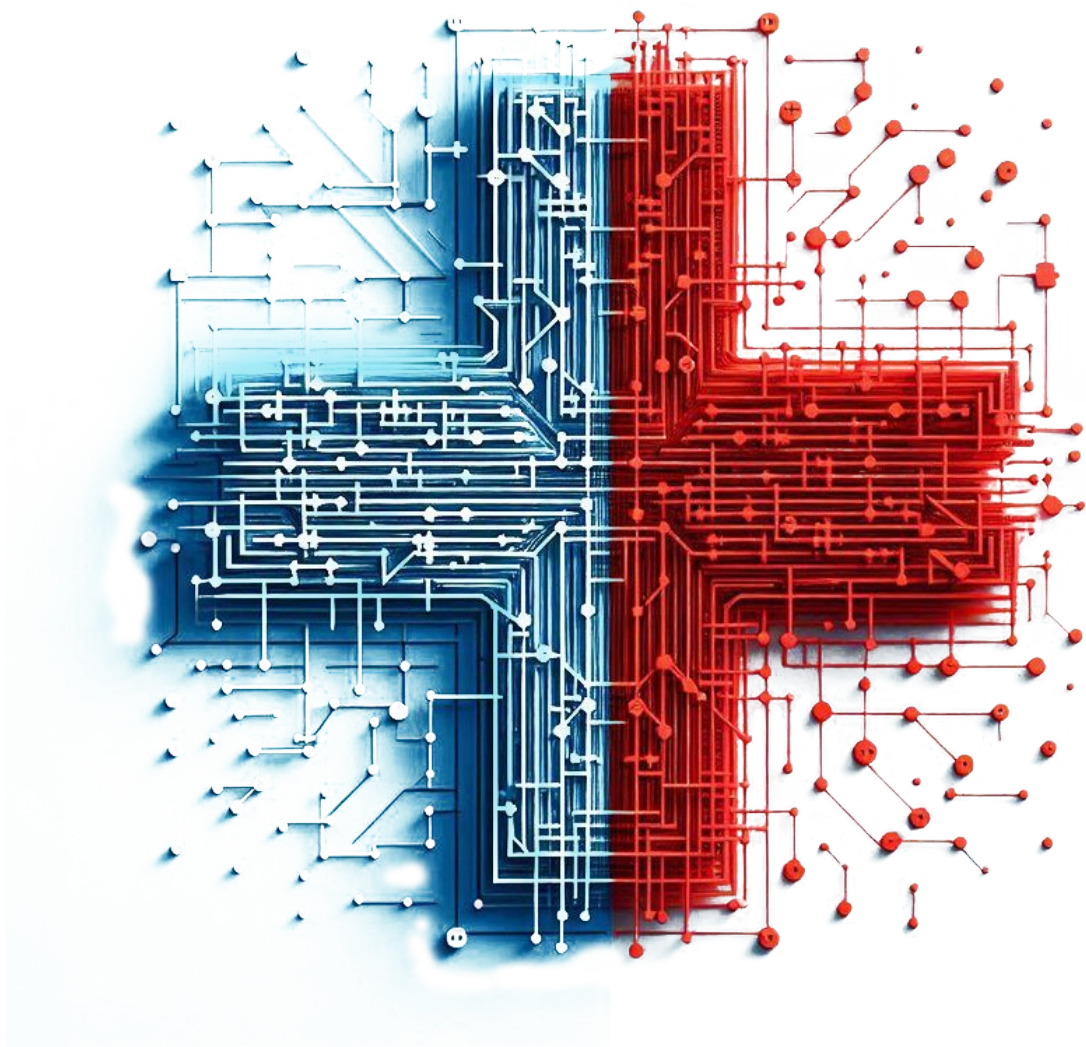
Legal necessity: Transparency and aligned to our values; responsible AI

R&D autonomy: Focus on what matters to us (multilinguality, compliance, local values)

Scientific integrity: Open data and reproducibility are foundational to science

Deployment control: On-premise, no vendor lock-in, data stays internal if you want

Teach the next generation: “What I cannot create, I do not understand”



APERTVS

EPFL

ETH zürich



CSCS

Fall 2025: Apertus 1.0 trained on “Alps”

Two sizes: 8B (16GB) and 70B (140GB)

for local and cloud deployments

~150k monthly downloads; 3.2M downloads total

Full Transparency

Open model to research and develop future AI systems with an OpenSource license

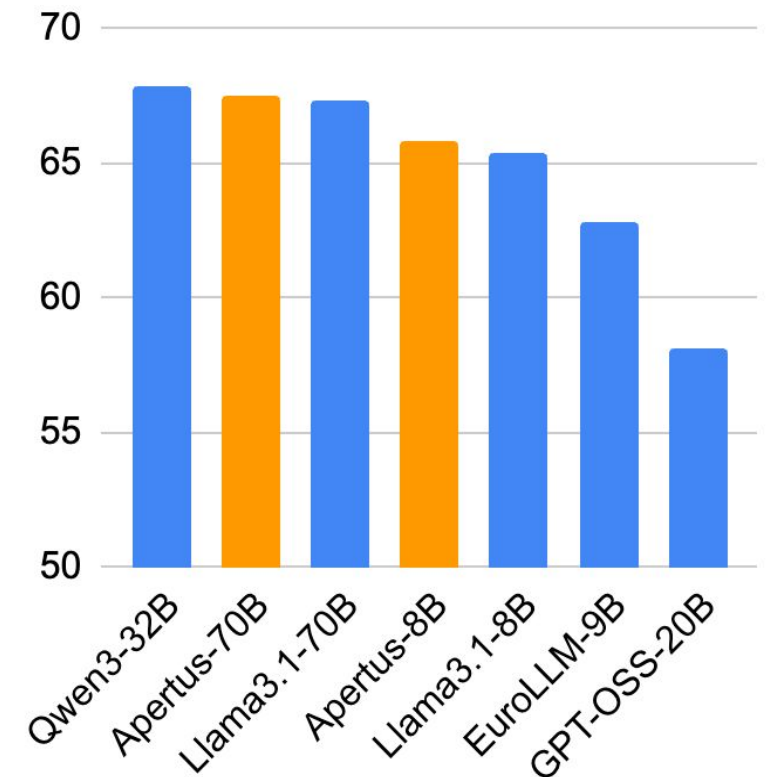
Trained on 15T tokens of text data

equivalent to reading 100M books or 60TB of raw text

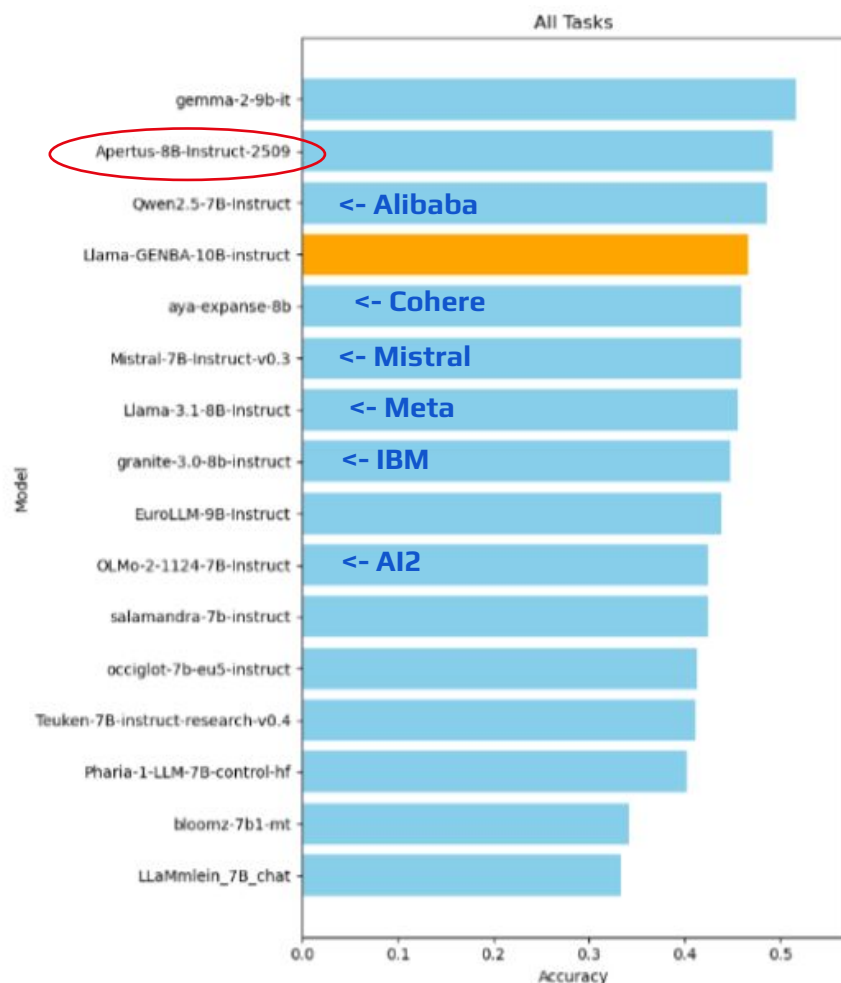
Required up to 4096 GPUs on Alps

Training on that scale for ~3 months

Strong English benchmark performance within class of fully-open models



Apertus 8B Belonged to the Best @ Release



[Third-party evaluation](#) of Apertus 8B by the Leibniz Supercomputing Center and Cerebras.

Model	Western Europe	Eastern Europe	Middle East	North Africa	Subsahar. Africa	Central Asia	South Asia	Southeast Asia	East Asia	Americas & Oceania	Avg.
7-10B Weight Class (IT)											
Qwen3 (8B)	80.6	79.1	74.2	66.8	56.3	70.2	76.0	83.0	82.4	52.0	75.1
Gemma 2 (9B)	78.1	76.1	70.5	64.8	43.7	65.0	71.1	79.5	75.0	53.0	70.4
Apertus (9B)	72.6	73.3	64.3	62.0	55.3	66.0	69.1	70.2	67.4	47.0	68.3
Qwen 2.5 (7B) <- Alibaba	72.4	69.0	69.8	59.8	57.5	59.0	64.2	76.8	74.4	55.0	67.6
Poro 2 (8B)	70.6	68.0	62.2	56.2	55.9	57.0	63.0	69.3	63.8	56.0	64.6
Aya Expanse (8B) <- Cohere	64.8	67.1	69.7	61.6	56.3	52.5	60.8	65.0	71.0	54.0	64.1
Krikri (8B)	67.9	66.6	60.7	58.8	55.9	52.2	60.6	67.3	65.6	49.0	63.2
Sailor2 (8B)	65.1	62.4	62.8	59.6	56.5	51.5	61.8	79.2	62.4	47.0	63.0
Llama 3.1 (8B) <- Meta	66.6	64.0	62.0	55.6	50.6	55.8	61.5	67.5	68.4	53.0	62.2
EXAONE 3.5 (7.8B)	64.4	62.1	61.0	54.0	54.8	53.8	62.2	61.5	66.4	56.0	61.7
Mixtral v0.1 8x7B <- Mistral	64.0	66.1	57.7	53.6	50.0	44.0	55.7	67.7	64.0	45.0	61.3
EuroLLM (9B)	66.2	67.3	62.8	60.2	52.3	48.0	52.8	57.7	65.4	50.0	60.7
Kanana 1.5 (8B)	64.3	61.6	58.6	53.8	54.4	52.0	58.5	65.7	70.6	51.0	60.6
SeaLLMs v3 (7B)	65.4	60.5	59.1	51.6	52.8	52.2	56.0	68.2	69.2	48.0	60.0
Mistral v0.3 (7B) <- Mistral	61.5	65.2	59.2	54.6	58.6	52.0	54.3	60.5	59.8	52.0	60.0
Command R (7B)	60.0	59.2	64.2	60.2	50.9	50.8	59.3	60.8	68.6	50.0	59.5
DeepSeek Qwen (7B) <- DeepSeek	59.8	59.8	52.9	53.0	48.8	46.0	54.3	56.2	60.8	44.0	56.2
Mistral v0.1 (7B)	55.4	57.4	51.9	51.6	51.1	48.8	50.0	54.2	52.4	52.0	53.9
VinaLLaMA (7B)	52.7	52.4	47.4	47.2	48.3	51.5	48.1	53.8	54.6	44.0	50.8
Tower v0.1 (7B)	52.5	52.5	48.9	50.8	52.1	50.8	47.0	49.3	52.2	47.0	50.8
Minerva v1.0 (7B)	48.8	49.3	46.6	47.0	47.7	48.2	46.8	47.8	44.8	42.0	47.7
Meltemi v1.5 (7B)	49.3	48.5	49.6	44.6	49.1	41.2	43.8	48.7	43.4	38.0	47.6
Salamandra (7B)	18.1	19.1	19.8	23.2	20.1	18.8	19.2	18.5	13.6	25.0	19.1

From Global PIQA: Evaluating Physical Commonsense Reasoning Across 100+ Languages and Cultures by Chang et al.

Unparalleled Openness

Training Scripts: github.com/orgs/swiss-ai

Deployment scripts

Training data pipelines

Evaluation harnesses

Model checkpoints: huggingface.co/swiss-ai

Technical report: arxiv.org/abs/2509.14233

**Full 119-page guide on how
to train a LLM from scratch**

APERTVS

DEMOCRATIZING OPEN AND COMPLIANT LLMs
FOR GLOBAL LANGUAGE ENVIRONMENTS

APERTUS V1 TECHNICAL REPORT

Project Apertus*

Core Team: Alejandro Hernández-Cano¹, Alexander Hägele¹, Allen Hao Huang¹, Angelika Romanou¹, Antoni-Joan Solergibert^{1,2}, Barna Pasztor², Bettina Messmer¹, Dhia Garbaya¹, Eduard Frank Durech^{1,2}, Ido Hakimi², Juan García Giraldo¹, Mete Ismayilzada¹, Negar Foroutan¹, Skander Moalla¹, Tiancheng Chen², Vinko Sabolčec¹, Yixuan Xu^{1,2}

Contributors: Michael Aerni², Badr AlKhamissi¹, Ines Altemir Marinas¹, Mohammad Hossein Amani¹, Matin AnsariPour¹, Ilia Badanin^{1,2}, Harold Benoit¹, Emanuela Boros¹, Nicholas Browning³, Fabian Bösch³, Maximilian Böther², Niklas Canova², Camille Challier¹, Clement Charmillot¹, Jonathan Coles³, Jan Deriu⁷, Arnout Devos², Lukas Drescher³, Daniil Dzenhaliov¹, Maud Ehrmann¹, Dongyang Fan¹, Simin Fan¹, Silin Gao¹, Miguel Gila³, Maria Grandury¹, Diba Hashemi¹, Alexander Hoyle², Jiaming Jiang¹, Mark Klein³, Andrei Kucharavy⁴, Anastasiia Kucherenko⁴, Frederike Lübeck², Roman Machacek⁹, Theofilos Manitaras³, Andreas Marfurt⁵, Kyle Matoba¹, Simon Matrenok¹, Henrique Mendonça³, Fawzi Roberto Mohamed³, Syrielle Montarior¹, Luca Mouchel¹, Sven Najem-Meyer¹, Jingwei Ni², Gennaro Oliva³, Matteo Pagliardini¹, Elia Palme³, Andrei Panferov⁶, Léo Paoletti¹, Marco Passerini³, Ivan Pavlov¹, Auguste Poiroux¹, Kaustubh Ponkshe¹, Nathan Ranchin¹, Javi Rando², Mathieu Sauser¹, Jakhongir Saydaliev¹, Muhammad Ali Sayfiddinov², Marian Schneider², Stefano Schuppli³, Marco Scialanga¹, Andrei Semenov¹, Kumar Shridhar², Raghav Singhal¹, Anna Sotnikova¹, Alexander Sternfeld⁴, Ayush Kumar Tarun¹, Paul Teiletche¹, Jannis Vamvas⁸, Xiaozhe Yao², Hao Zhao¹

Advisors: Alexander Ilic², Ana Klimovic², Andreas Krause², Caglar Gulcehre¹, David Rosenthal¹⁰, Elliott Ash², Florian Tramèr², Joost VandeVondele³, Livio Veraldi¹⁰, Martin Rajman¹, Thomas Schulthess³, Torsten Hoefler²

Leads: Antoine Bosselut¹, Martin Jaggi¹, Imanol Schlag²

Global Representation

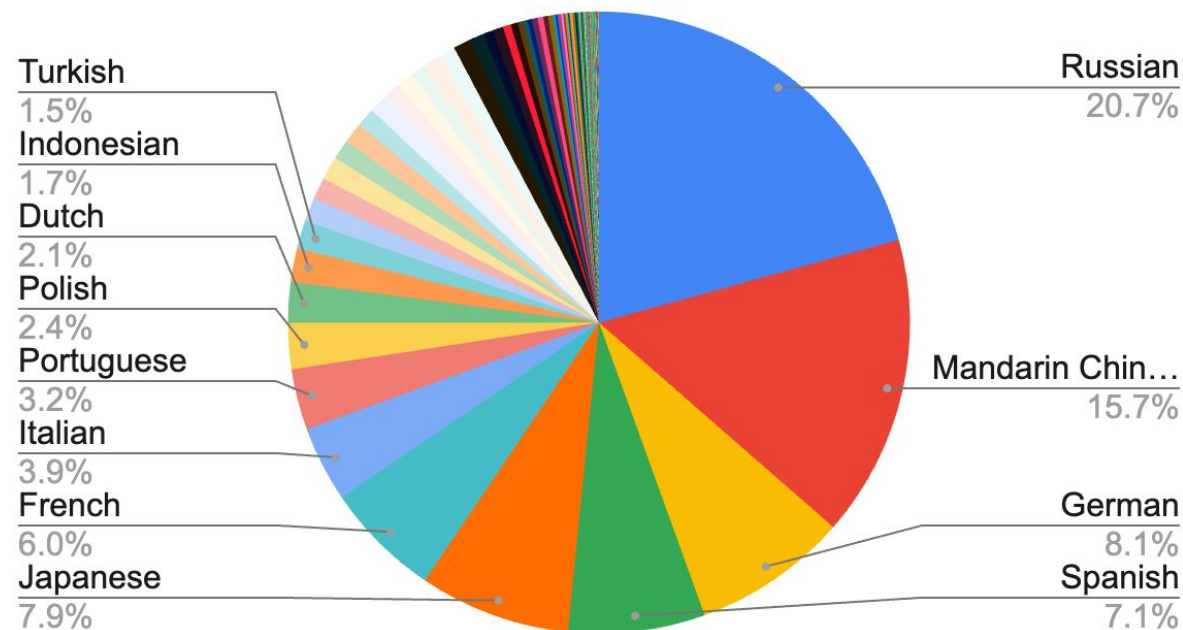
1000+ languages in training data.

Data reflects **the web distribution** of languages on the web:

~60% **English** web data

~40% **non-English** web data
(many low-resource languages)

Conversational in dozens of languages.



The Different Training Stages



Pretraining

Compute
Very High ●●●●●●●●

- Read enormous amounts of text from books and the web
- Pick up grammar, facts, and patterns of language
- Build a base of general knowledge

Example

The capital of France is →
Paris



Supervised Fine-Tuning (SFT)

Compute
Very Low ●

- Learn to follow instructions
- Practise giving helpful answers from human-written examples
- Move from raw text to a chat-like style

Example

How do I bake a chocolate cake? →
Recipe: 1) Preheat oven to 175°C...



RL from Verifiable Rewards (RLVR)

Compute
Medium ●●●●

- Practise tasks with clear right or wrong answers
- Try many attempts and keep what works
- Get better at reasoning, maths, and code

Example

Solve: $3x + 2 = 11$ →
 $x = 13$ $x = 3$ ✓
 $x = 5$ $x = 9$

new in
Apertus 1.5



Preference Training

Compute
Very Low ●

- Learn which answers humans prefer
- Become safer and more helpful
- Avoid harmful or misleading replies

Example

How do I deal with stress? →
Try meditation or exercise ✓
Just drink alcohol ✗

The Apertus Charter

Apertus behaviour is trained **without direct human feedback**

A diversity of strong open-weight **AI models judge Apertus outputs** on the basis of the Charter

The charter reflects our current view on how AI models ought to behave

Version 1.0
August 2025

PREAMBLE

This charter sets forth principles for the alignment of artificial intelligence systems developed under the Swiss AI Initiative. Rooted in Switzerland's constitutional values, democratic traditions, and shared commitment to human dignity, these principles are designed to translate abstract values into concrete alignment criteria for training large language models (LLMs). As AI capabilities advance and our understanding of alignment matures, this charter will adapt through participatory refinement, ensuring that our approach remains both principled and responsive to social and technological change.

LIST OF ARTICLES

1. **Response Quality** — Writing clear, accurate, and useful responses.
2. **Knowledge and Reasoning Standards** — Using verified facts and sound reasoning.
3. **Respectful Communication** — Treating people with courtesy, fairness, and accessibility.
4. **Preventing Harm** — Protecting safety and refusing harmful requests.
5. **Resolving Value Conflicts** — Handling trade-offs openly and preserving principles.
6. **Professional Competence Boundaries** — Educating without giving licensed advice.
7. **Collective Decision-Making** — Supporting fair and constructive group decisions.
8. **Autonomy and Personal Boundaries** — Respecting choice, privacy, and clear limits.
9. **Long-term Orientation and Sustainability** — Considering long-term impacts and risks.
10. **Human Agency** — Keeping humans in control and independent.
11. **AI Identity and Limits** — Being clear about what the AI is and is not.

CHARTER TEXT

1. Response Quality. The AI should ensure that every response is helpful, harmless, and honest [1.1]. Accuracy, completeness, and usefulness must always take priority, with factual correctness placed above style or polish [1.2]. Each response should fully address the user's question with a level of detail and complexity that matches the scope of the request, keeping explanations concise and proportionate [1.3]. Responses should provide guidance that helps users solve their problems or

Key Enabler: ALPS Research Infrastructure

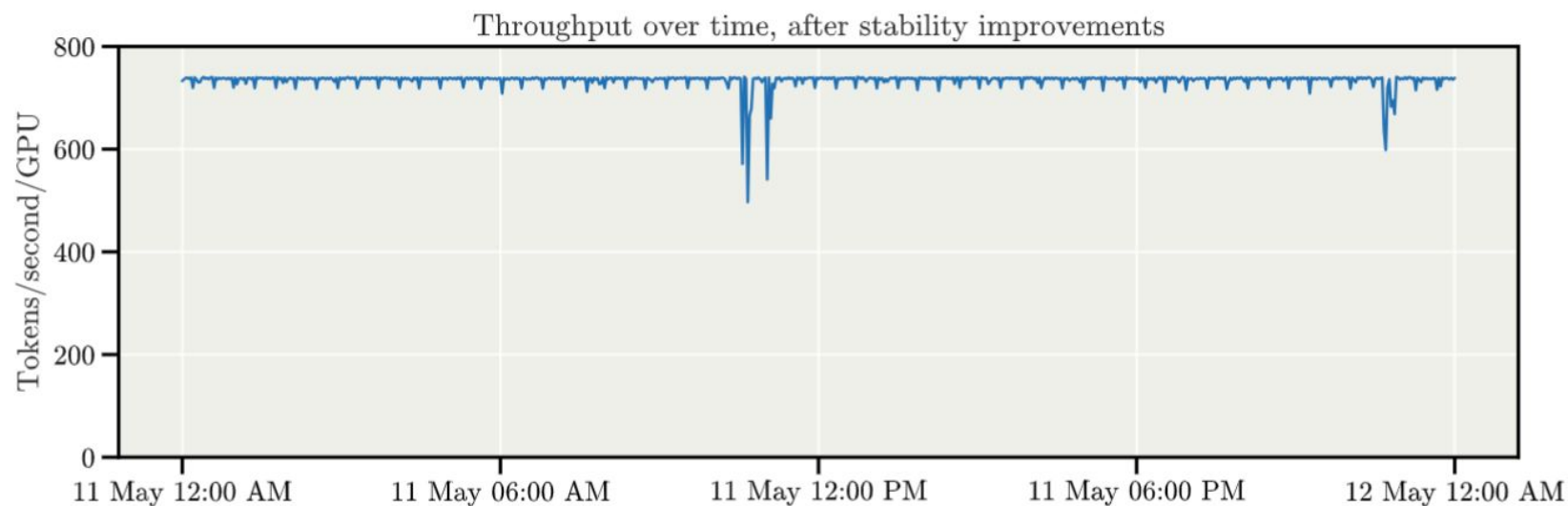
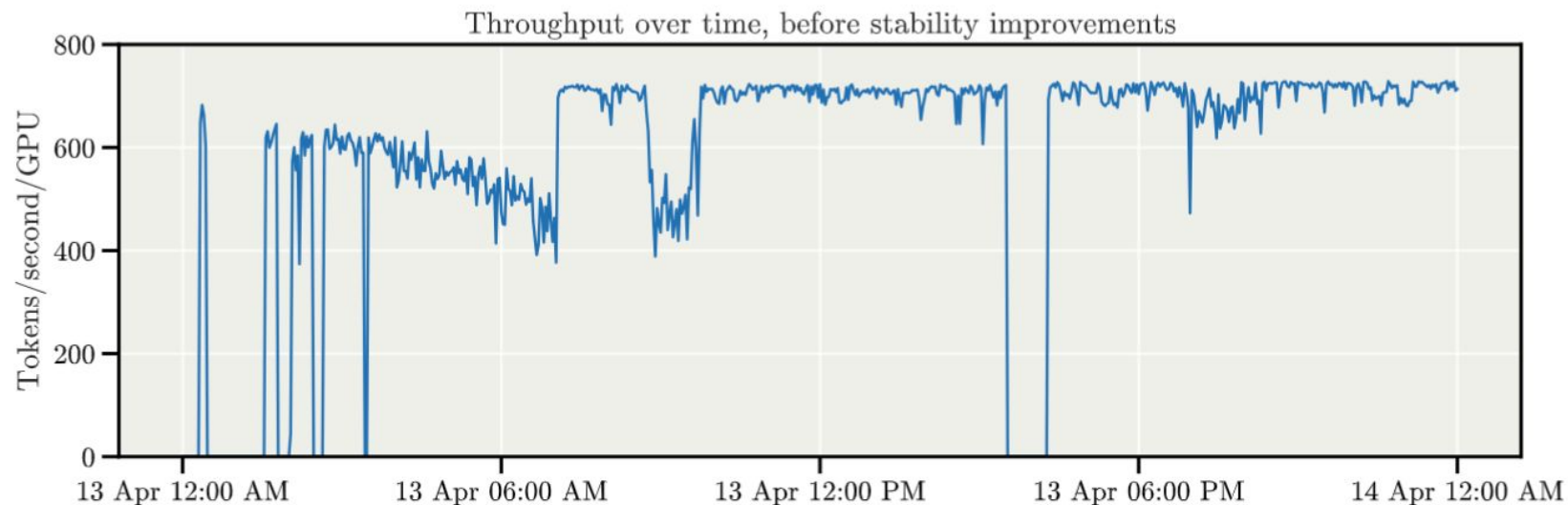
ALPS: one of Europe's largest (11,000 Nvidia GH200 GPUs) and efficient computing infrastructures

Built and maintained by a large team at CSCS, in close collaboration with HPE and NVIDIA

Full-fledged AI platform now operational



Training at Scale Is Hard



Energy and Efficiency

4096 GPUs for 3 months for pre-training the 70B model.

Requires **the power of a locomotive** (at 50% peak).

Energy usage for training Apertus equivalent to that of four 30-year-old Swiss.

Low emissions: hydro-powered & lake-water cooled



Power envelope full Alps:
10MW or about two locomotives
at full-speed.



Pre-training energy use: 5GWh or
about four 30-year-old Swiss at
4750W per person

July 2026: Apertus 1.5

- native image understanding alongside text
- experimental audio input with focus on spoken speech
- thinking model for stronger performance on reasoning problems
- 4x longer context of 262,144 tokens
- improved instruction following
- improved tool-use



Describe the image please

apertus-ai/Apertus-v1.5-70B

This is a political propaganda poster from the late 19th or early 20th century, likely from Switzerland (based on the German text and the Swiss cross emblem). The poster contrasts two political ideologies: **Majorz** (plurality voting) and **Proporz** (proportional representation).

Key Elements of the Image:

1. Title:

- At the top, the text reads: "**Gerechtigkeit erhöht ein Volk!**" which translates to "**Justice elevates a people!**" This sets the moral tone of the poster.

2. Left Panel (Majorz - Plurality Vote):

- **Visuals:**
 - A large, oppressive figure wearing a top hat is depicted greedily stuffing a sausage into his mouth from a plate. The sausage is labeled "**Majorz**" on the tablecloth.
 - The tablecloth also features a bird, possibly symbolizing freedom or the people, which appears subdued or neglected.
 - In the background, a woman in traditional Swiss attire (with a wreath of leaves) covers her face in distress, while children stand nearby looking sad and neglected.
- **Interpretation:**
 - The "Majorz" figure symbolizes a dominant political force or party that hoards the

July 2026: Apertus 1.5

- Continued Pretraining from Apertus 1.0
 - 8B: 15T -> 19T
 - 70B: 15T -> 17T
- Completely new post-training
- fully open AI
 - open weights
 - open data
 - open methods
 - open values (Apertus Charter)

apertus-ai/Apertus-v1.5-70B ▾ +

does taiwan belong to china?

oi apertus-ai/Apertus-v1.5-70B

The question of whether Taiwan belongs to China is complex and contentious, involving historical, political, legal, and cultural dimensions. Here's a balanced overview of the different perspectives:

Chinese Perspective:

- **Historical Claim:** The People's Republic of China (PRC) and the Republic of China (ROC, which governs Taiwan) both claim that Taiwan has been an integral part of Chinese territory since ancient times. The PRC specifically asserts that Taiwan has been part of China since the Qing dynasty and was ceded to Japan in 1895 following the First Sino-Japanese War but was returned to China after World War II.
- **One-China Policy:** The PRC adheres to the "One-China Principle," which considers Taiwan to be a part of China, with the PRC being the sole legitimate government representing the whole country, including Taiwan. This principle is the basis for the PRC's demand for eventual reunification of Taiwan with mainland China, ideally on terms acceptable to the PRC.

Taiwanese Perspective:

- **Self-Governance:** The Republic of China (Taiwan) operates as a separate, self-governing entity with its own government, military, currency, and passport. Taiwan has not been under the control of the PRC since the end of the Chinese Civil War in 1949, when the ROC government retreated to Taiwan.
- **Distinct Identity:** Many in Taiwan identify as Taiwanese rather than Chinese, emphasizing a distinct Taiwanese identity and culture, which has grown stronger over the decades. Polls often show that a majority of Taiwanese prefer maintaining the current status quo of de facto independence rather than unification with the PRC immediately.
- **Political Status:** The political status of Taiwan is ambiguous. The ROC government initially claimed to be the legitimate government of all of China but has since moved away from that stance in practice, though it still retains the name and some symbols of the former Chinese government. Taiwan functions as a democratic state with no official diplomatic recognition by most countries due to pressure from the PRC adhering to the One-China Policy.

International Perspective:

- **Diplomatic Recognition:** Most countries, including the United States, Japan, and many European nations, maintain "One-China Policies" in their diplomatic relations, acknowledging the PRC as the sole government of China while having unofficial relations with Taiwan through

Try it yourself

serving.swissai.cscs.ch: free if you have ethz/epfl email and VPN

publicai.co: free third party service

oss.zuericitygpt.ch: free third party service RAG based on the website of the city of Zürich

www.infomaniak.com/en/hosting/ai-services/open-source-models:
API input CHF 0.7/1M tokens; CHF 2.5/1M tokens

(more soon)

Coming Soon: Apertus 2.0

Scaling up: What is the largest model we can train?

- ~8-10x larger
- stronger performance
- same deployment costs

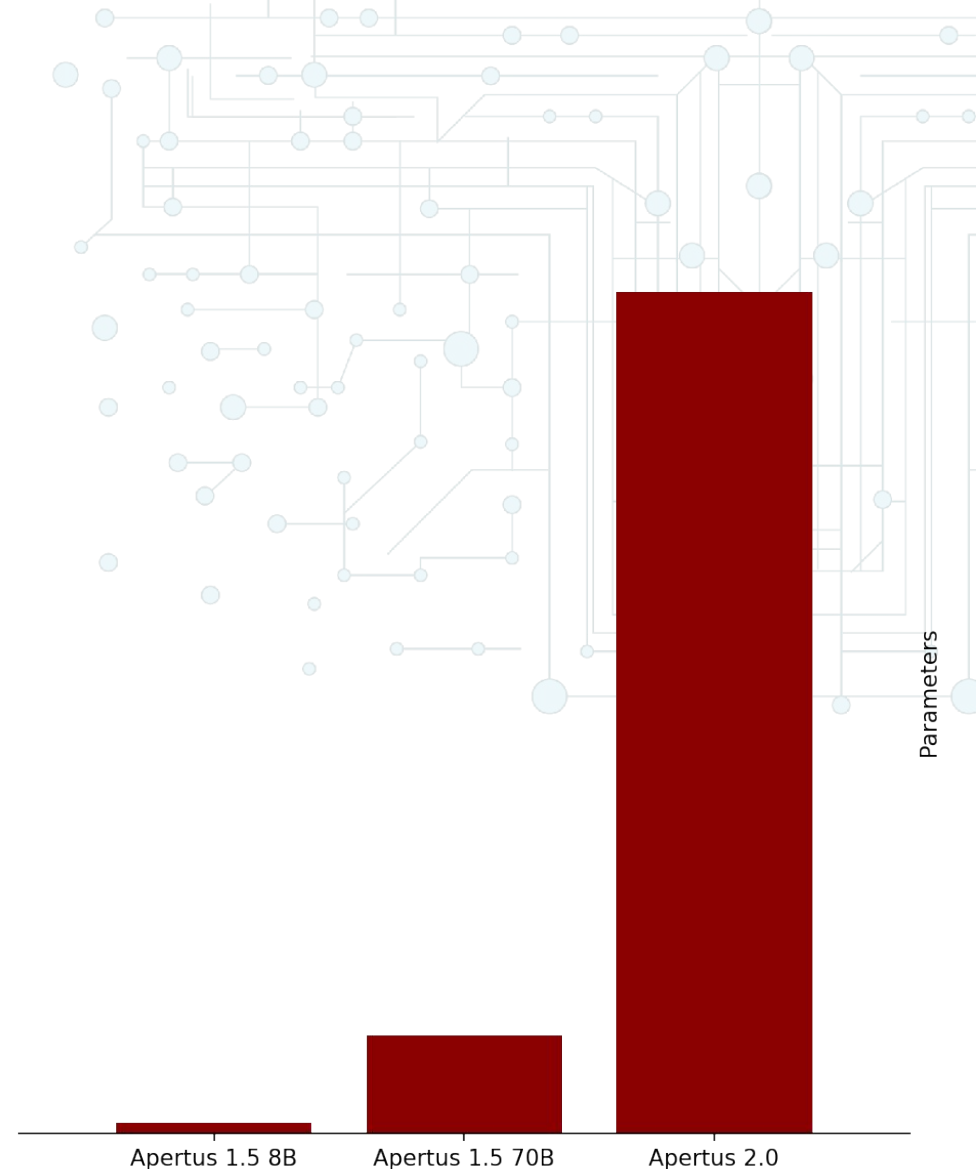
Native Multimodality (Image & Audio Inputs)

Leading fully open models

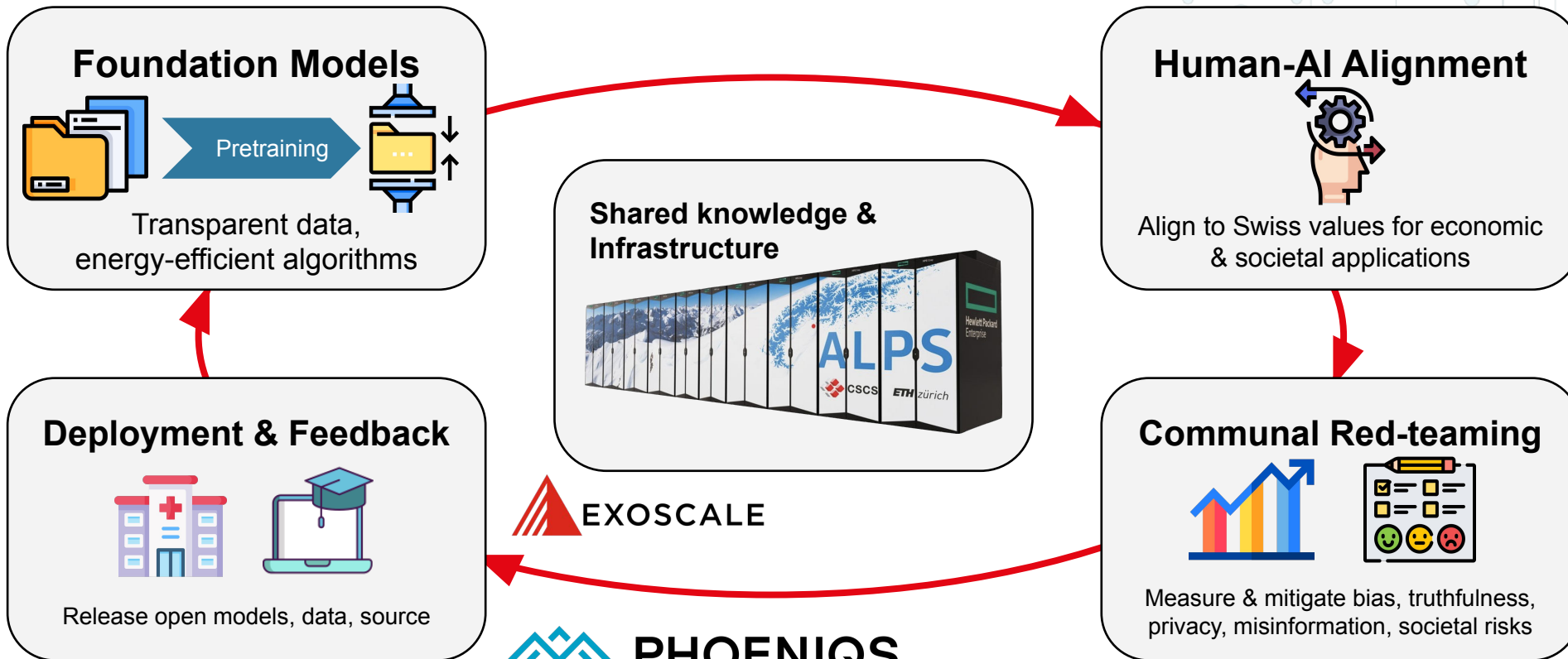
transparent, responsible, inclusive

Focus Domains

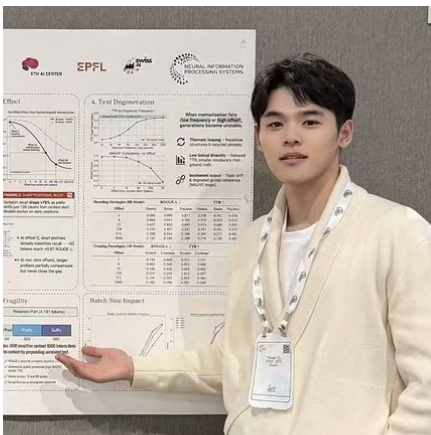
health, education, justice, and advanced reasoning



Release Cycle



The Apertus Engineers



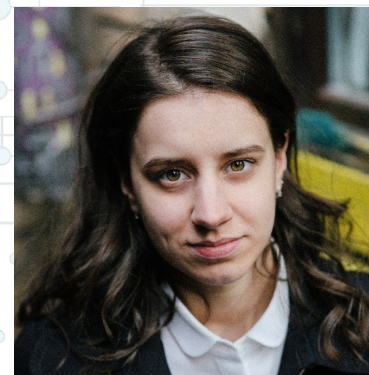
post-training/multimodality



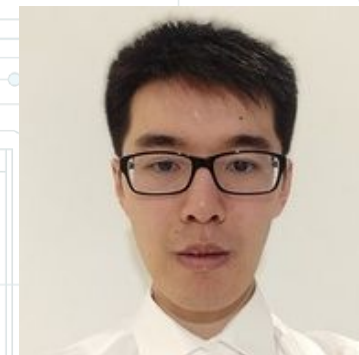
data/compliance



program manager



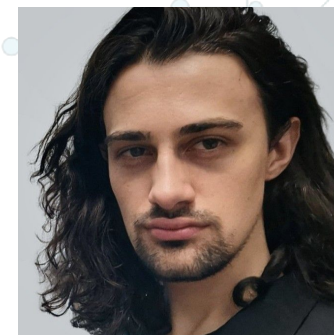
posttraining/evaluations



pretraining



infrastructure/posttraining



posttraining/infrastructure



pretrain/post-train



pretraining/multimodality



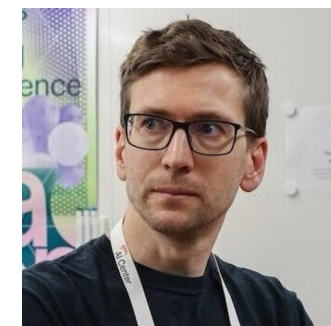
data



posttraining/planning



data/evaluations

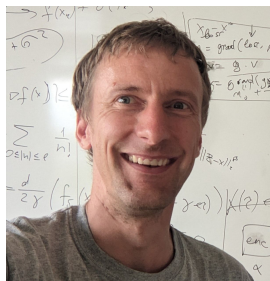


community/evaluations

Some of the Apertus Academics ...



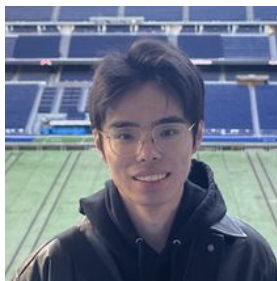
co-lead (prof)



co-lead (prof)



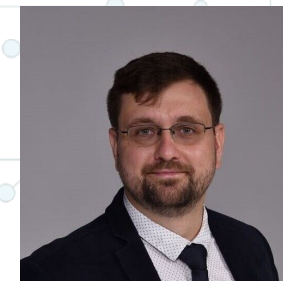
optimisation (PhD)



architecture (MSc)



pretrain safety (PhD)



safety (prof)



infrastructure (MSc)



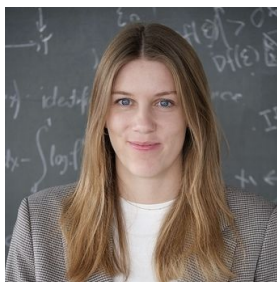
interpretability
(Postdoc)



serving (MSc)



alignment (MSc)



evals (PhD)



data (Postdoc)



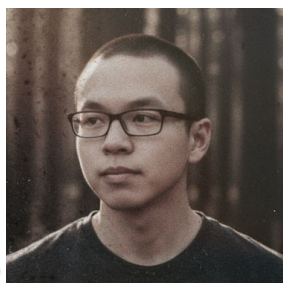
RL (MSc)



Scaling (PhD)



alignment (MSc)



audio (MSc)



alignment
(Postdoc)



RL (MSc)



evals (PhD)



code (PhD)



vocabulary (PhD)

AI Is Changing the World

Intelligence is becoming a commodity: abundant, cheap, everywhere

Outsourcing **critical infrastructure** brings risks:

- **Data:** whoever serves the model has unprecedented access
- **Behaviour:** whoever trains the model decides how it will act

We are developing the expertise and infrastructure to build, evaluate, and deploy such AI on our own terms.

Thank You!

Find more info on
our website: apertus-ai.org



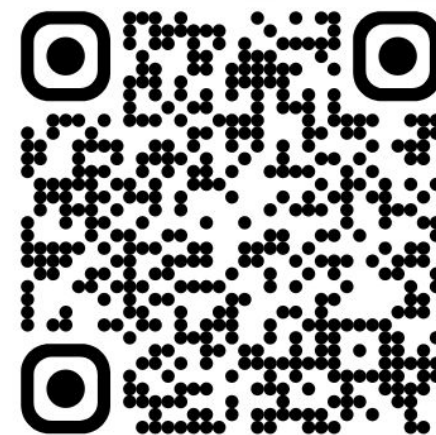
sign-up form for student
opportunity waitlist

Speaker
Imanol Schlag, PhD
AI Research Scientist
Apertus Co-Lead
ETH Zürich, ETH AI Center



ETH AI CENTER

APERTVS



sign-up for our newsletter
for latest updates